Vision of the Department

To be a well-known centre for pursuing computer education through innovative pedagogy, value-based education and industry collaboration.

Mission of the Department

To establish learning ambience for ushering in computer engineering professionals in core and multidisciplinary area by developing Problem-solving skills through emerging technologies.

Session 2025-2026

Vision: Dream of where you want.	**Mission:** Means to achieve Vision

Program Educational Objectives of the program (PEO): (broad statements that describe the professional and career accomplishments)

PEO1	**Preparation**	**P: Preparation**	**Pep-CL abbreviation pronounce as Pep-si-lL easy to recall**
PEO2	**Core Competence**	**E: Environment (Learning Environment)**	
PEO3	**Breadth**	**P: Professionalism**	
PEO4	**Professionalism**	**C: Core Competence**	
PEO5	**Learning Environment**	**L: Breadth (Learning in diverse areas)**	

Program Outcomes (PO): (statements that describe what a student should be able to do and know by the end of a program)

Keywords of POs:
Engineering knowledge, Problem analysis, Design/development of solutions, Conduct Investigations of Complex Problems, Engineering Tool Usage, The Engineer and The World, Ethics, Individual and Collaborative Team work, Communication, Project Management and Finance, Life-Long Learning

PSO Keywords: Cutting edge technologies, Research

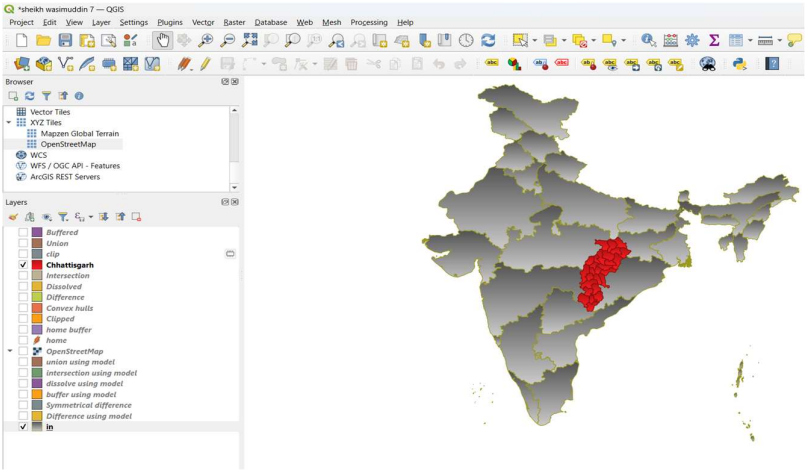
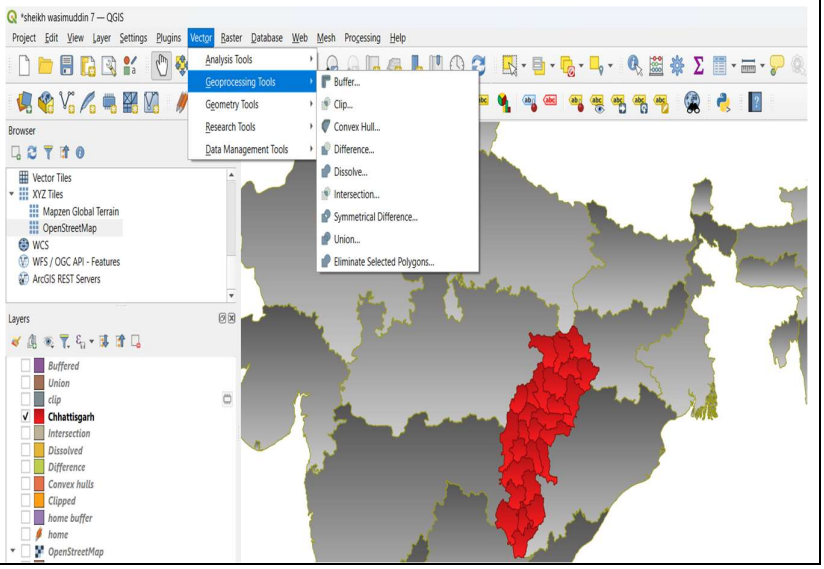
"I am an engineer, and I know how to apply engineering knowledge to investigate, analyse and design solutions to complex problems using tools for entire world following all ethics in a collaborative way with proper management skills throughout my life." *to contribute to the development of cutting-edge technologies and Research*.

Integrity: I will adhere to the Laboratory Code of Conduct and ethics in its entirety.

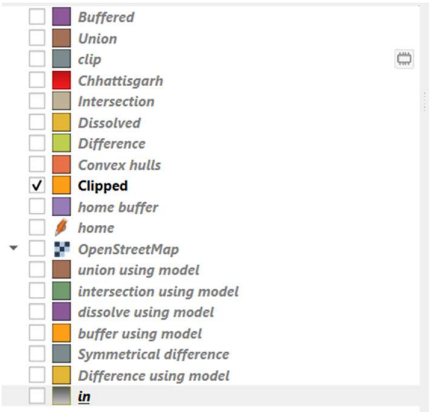
Name and Signature of Student and Date
(Signature and Date in Handwritten)

Session	**2025-26 (ODD)**	**Course Name**	**PE-I - Geo-Intelligence for Smart IoT Devices Lab**
Semester	**5**	**Course Code**	**23IOT1523**
Roll No	**64**	**Name of Student**	**Sheikh Wasimuddin**

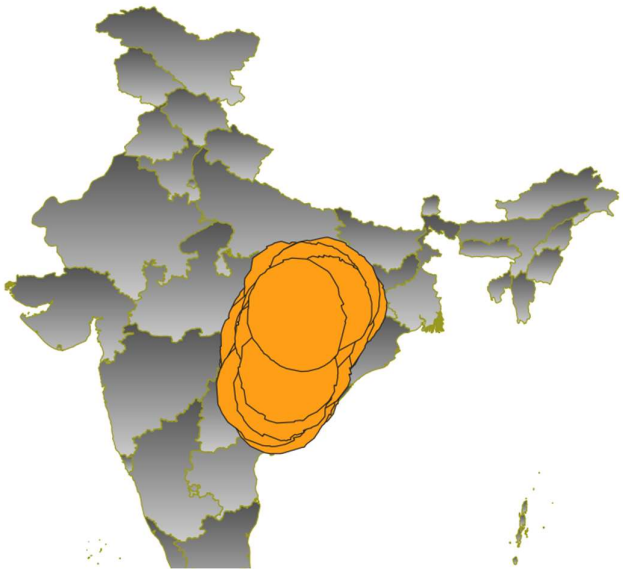
Practical Number	**07**
Course Outcome	Apply and demonstrate the use of proprietary and open-source GIS tools (e.g., QGIS) for creating, visualizing, and managing spatial datasets.
Aim	Implement Geo-processing Tools(Clip, Buffer, Difference, Union, Intersection and Dissolve) on various vector data.
Problem Definition	To perform spatial analysis on the Chhattisgarh shapefile by applying geo-processing tools such as Clip, Buffer, Difference, Union, Intersection, and Dissolve for effective data refinement and visualization
Theory (100 words)	Geo-processing tools are essential components of spatial analysis in Geographic Information Systems (GIS), enabling the manipulation and analysis of vector data for meaningful insights and visualization. In this study, various geo-processing tools such as Clip, Buffer, Difference, Union, Intersection, and Dissolve are applied to the *Chhattisgarh shapefile* to refine and analyze spatial features effectively. The Clip tool extracts features from an input layer that fall within the boundary of another layer, allowing focus on a specific area of interest within Chhattisgarh. The Buffer tool creates zones around spatial features (points, lines, or polygons) at a specified distance, which is useful for proximity analysis, such as identifying regions near rivers or roads. The Difference tool subtracts the overlapping areas of one layer from another, helping in identifying non-overlapping or exclusive regions. The Union tool merges multiple datasets to form a new layer containing all spatial features and their attributes from both layers. The Intersection tool extracts overlapping areas between two layers, revealing the common spatial regions that meet specific conditions. Finally, the Dissolve tool generalizes data by merging adjacent polygons with similar attributes into a single feature, reducing data complexity and enhancing visualization. Together, these geo-processing operations facilitate efficient spatial data refinement, pattern recognition, and decision-making for geographic studies and planning in the Chhattisgarh region.

Procedure and Execution (100 Words)	Implementation Steps: 1. Download India and keralas Shapefile 2. Import it as vector layers 3. Go to Vector -> Geoprocessing Tools 4. Select inputs as vector and algorithm as clipped, Buffer, Difference, Dissolve, Intersection and Union 5. Select Input .shp 6. Your output Shape file will be visible on Qgis. 1.  2. 

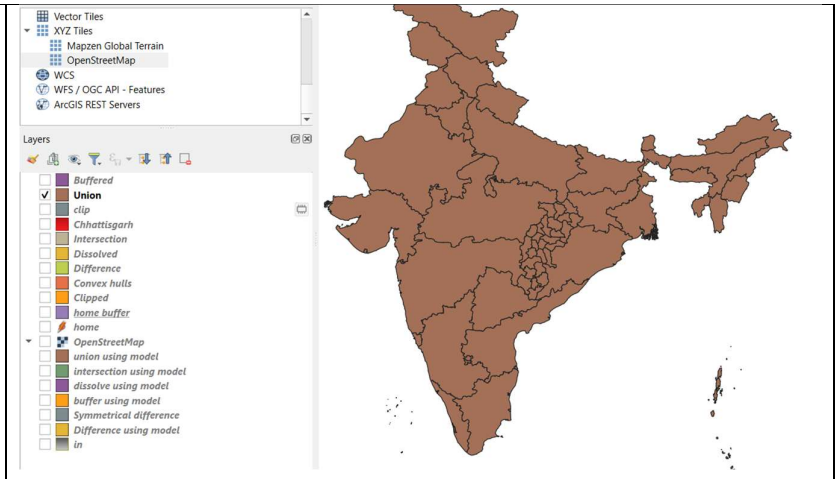
3. CLIPPED



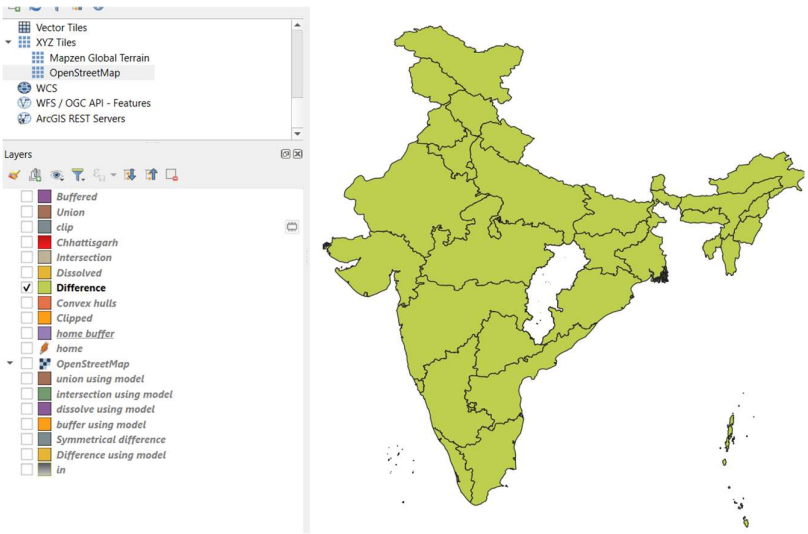
4. BUFFER



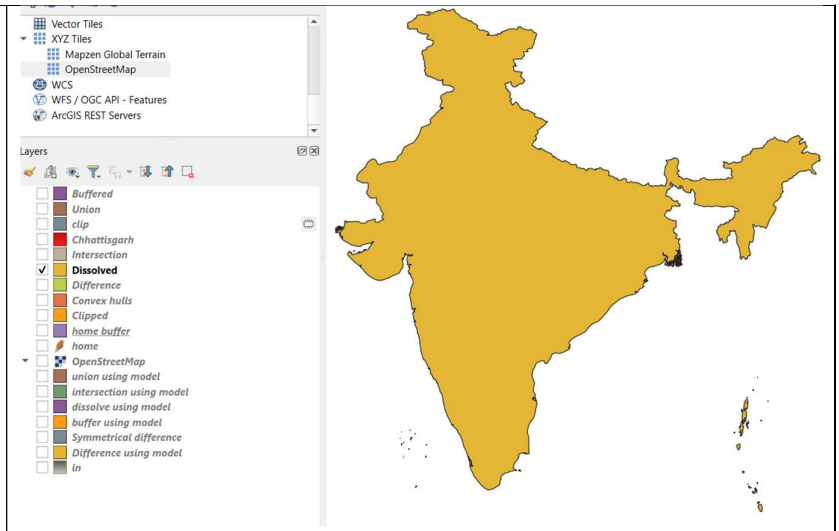
5. UNION



6. DIFFERENCE



7. DISSOLVED

	 8. INTERSECTION
Output Analysis	After applying various geo-processing tools on the *Chhattisgarh shapefile*, distinct spatial layers were generated, each representing different analytical outcomes. The Clip operation successfully extracted the area of interest from the larger dataset, isolating Chhattisgarh and enabling focused analysis within its boundaries. The Buffer tool produced surrounding zones around key spatial features such as roads, rivers, or urban centers, allowing assessment of influence areas and accessibility patterns. Through the Difference operation, non-overlapping regions were clearly identified, highlighting areas excluded from specific spatial features or zones. The Union tool integrated multiple thematic

	layers into a comprehensive dataset, preserving all spatial and attribute information, which facilitated holistic visualization and analysis. The Intersection operation effectively delineated the overlapping zones between datasets, such as areas where land use, vegetation, and administrative boundaries coincide, offering valuable insights into spatial relationships. Finally, the Dissolve operation simplified the dataset by merging polygons sharing common attributes, resulting in cleaner and more interpretable maps. Overall, these geo-processing tools enhanced spatial understanding, data accuracy, and map readability, enabling effective visualization and decision-making for regional planning and geographic analysis in Chhattisgarh.
Conclusion	The implementation of geo-processing tools such as Clip, Buffer, Difference, Union, Intersection, and Dissolve on the *Chhattisgarh shapefile* proved highly effective in refining spatial data and enhancing geographic visualization. Each tool played a significant role in manipulating vector datasets to extract, combine, and simplify spatial features based on specific analytical needs. The process not only improved the accuracy and clarity of spatial information but also provided valuable insights into spatial relationships and regional patterns within Chhattisgarh. These operations demonstrated the power of GIS in supporting data-driven decision-making, planning, and resource management. Overall, the application of geo-processing tools contributed to a deeper understanding of the region's spatial characteristics and established a strong foundation for advanced spatial analysis and thematic mapping.
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